Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Zachary Sullivan	zsull@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

A game that will have the user guess which math operation is being used in an equation

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are existing learning websites such as IXL and Cengage Learning, but this will be more gamified, with a high score streak tracker.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I would want this implemented to be used as a fun learning device to help students improve their quick math operations.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The ultimate goal is for the user to have a fun experience while improving their quick math skills.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

A numerical operation or value that will complete the equation.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

It will compare the input against the expected numerical result to determine if points will be awarded.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get back a “Correct answer” or “Incorrect Answer” screen, and if they get the question correct, they will have a point added to their score.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Score, Streak Counter, Question, Operation

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need for Operation: operation type, result of the expression, boolean if the user answered correctly
Details I will ignore for Operation: Explanation of operation, decimal result(all answers will be rounded if needed), result of last operation

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

Creating a streak counter for the consecutive correct answers that the user gets correctly, and providing a lengthy amount of question prompts.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will ensure that I clearly understand what the LLM outputs by asking it to fully break down and justify each part of its code. If something does not make sense, I will ask it to revise the code to better align with my goals.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

To prove to myself that I can create a fun and engaging learning game to help others learn math operations.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple web game that helps students learn math operations in an engaging way. What are 3 limitations of this idea and what are the 3 most important componets to include?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The AI suggested a reward system to keep the learner engaged.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

The adaptive difficulty is too complex for the scope of this project. Is there a simpler alternative to this?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

It has introduced the idea of difficulty levels which can help increase engagement to learners that are both beginners and advanced in mathematics.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will go to the TA office hours to ask for advice on how to continue my project.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

Identify the error by pasting it in Thonny and creating a new smaller file with just the erroneous part, which I will then use the debug feature to identify the issue. If I am not able to debug by myself, I will go to the TA's office to ask for advice.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Q: I want to build a simple web game that helps students learn math operations in an engaging way. What are 3 limitations of this idea and what are the 3 most important componets to include?

A: For a **simple math operations web game**, here are the most important considerations:

3 Limitations

1. **Students may lose interest quickly**
    - If the game only asks math questions repeatedly, it can feel like a digital worksheet instead of a game.
2. **Different skill levels**
    - Some students may find the problems too easy while others find them too difficult, making it hard to keep everyone engaged.
3. **Learning vs. entertainment balance**
    - Adding too many game features can distract from learning, while too little gameplay can

make the experience boring.

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3 Most Important Components

1. **Adaptive Difficulty**
    - Increase or decrease problem difficulty based on the student's performance.
    - Example: Start with single-digit addition and gradually introduce larger numbers or mixed operations.
2. **Immediate Feedback**
    - Tell students right away whether an answer is correct.
    - Include explanations or hints for incorrect answers to reinforce learning.
3. **Reward and Progress System**
    - Points, badges, levels, streaks, or unlocking new characters.
    - Gives students a reason to keep practicing.

Simple Game Concept

Math Quest

- Players defeat monsters by solving math problems.
- Correct answer = attack monster.
- Incorrect answer = lose a small amount of health and receive a hint.
- Earn coins to unlock new avatars and backgrounds.
- Difficulty increases as levels progress.

This keeps the focus on learning while adding enough game elements to make practice feel rewarding rather than repetitive.

Q: the adaptive difficulty is too complex for the scope of this project. Is there a simpler alternative to this?

A: Yes. For a small project, I'd replace **adaptive difficulty** with one of these simpler features:

Option 1: Difficulty Selection (Recommended)

Let the player choose:

- Easy (single-digit problems)
- Medium (double-digit problems)

- Hard (mixed operations)

Why it's good: Very easy to implement and still gives students control over challenge level.

Option 2: Level Progression

Create fixed levels:

- Level 1: Addition
- Level 2: Subtraction
- Level 3: Multiplication
- Level 4: Division

Players unlock the next level after getting a certain score.

Why it's good: Feels like game progression without needing algorithms to adjust difficulty.

Option 3: Timed Challenge

Keep all questions at roughly the same difficulty but add:

- 30-second rounds
- Bonus points for answering quickly

Why it's good: Creates excitement without changing the math itself.

Revised Top 3 Components

For a simple semester project, I would use:

1. **Immediate Feedback**
    - Correct/incorrect messages
    - Show the right answer when wrong
2. **Points and Rewards**
    - Score counter
    - Badges, stars, or coins
3. **Levels or Difficulty Selection**

- Easy/Medium/Hard buttons or unlockable levels

These three components are straightforward to build and will likely give the biggest increase in engagement for the amount of work required.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.